

THE GEORGE WASHINGTON UNIVERSITY

WASHINGTON, DC

Data Science Program
Department of Data Science
DATS 6501 - Capstone Project - Fall 2026

Post-Training Analysis & Novelty Detection Using Neural Network Self-Organizing Maps (NNSOM)

PHASE 2 REPORT

NNSOM Training, Cluster and Error Analysis

Capstone - Group 5

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1. SOM 15×15 Overview

This phase applies a 15×15 Self-Organizing Map (SOM) to the 84-dimensional `fc2` representations extracted from the pretrained MNIST CNN. The SOM was trained with seed 42 on 49,000 training embeddings and validated on 10,500 held-out validation embeddings. The objective is to examine whether topology-preserving organization of the learned representation reveals class structure, density patterns, and localized validation weaknesses Kohonen 1990; Jafari n.d.

2. SOM TRAINING

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3. CLUSTER ANALYSIS

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4. ERROR ANALYSIS

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5. Leakage Control

The SOM work uses the training and validation splits only:

- The SOM was trained on the 49,000 training embeddings. The scaler that maps the embeddings to $[-1, 1]$ was fitted on the training embeddings only and then applied unchanged to the validation embeddings.
- SOM settings were compared on validation data only (the configuration sets `use_test_for_selection: false`). The 15×15 grid follows the instructor's request; its selection history is recorded in the project's deviation log.
- The maps, purity, label-shuffle control and error maps in this report use validation images. The test embeddings are never loaded: the model's metrics file records that the test split was not evaluated, and the analysis code refuses to run otherwise.
- The CNN that produced the embeddings used the same validation split to choose its checkpoint, so validation is not entirely untouched by model development; the test split remains unused for the final evaluation.

6. Discussion

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7. Conclusion and Next Phase

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References

Kohonen, T. (1990). The Self-Organizing Map. *Proceedings of the IEEE* 78.9, pp. 1464–1480.

Jafari, A. (n.d.). NNSOM: Neural Network Self-Organizing Maps in Python. The George Washington University. URL: https://amir-jafari.github.io/SOM/.